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Indian Institute of Technology Hyderabad

Concentration Inequalities (AI2200)

Lecture 01

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Markov's Inequality

Theorem 1 (Markov's Inequality)

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Let X be a non-negative random variable with $\mathbb{E}[X] < +\infty$. Then, for any fixed $\alpha > 0$,

$$\mathbb{P}(\{X \geq \alpha\}) \leq \frac{\mathbb{E}[X]}{\alpha}.$$

Remarks:

- Markov's inequality applies only to non-negative random variables
- Although the inequality technically holds for any $\alpha > 0$, it is meaningful only for $\alpha > \mathbb{E}[X]$
- **Proof of Theorem 1:** For any fixed $\alpha > 0$,

$$\mathbb{E}[X] = \mathbb{E}[X \mathbf{1}_{\{X \geq \alpha\}}] + \mathbb{E}[X \mathbf{1}_{\{X < \alpha\}}] \geq \mathbb{E}[X \mathbf{1}_{\{X \geq \alpha\}}] \geq \alpha \mathbb{P}(\{X \geq \alpha\}).$$

Rearranging terms, we arrive at the desired relation

- **Question:** For a fixed $\alpha > 0$, what distribution of X achieves equality in Markov's inequality?

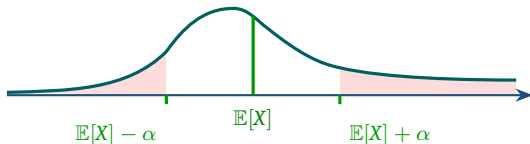
Chebyshev's Inequality

Theorem 2 (Chebyshev's Inequality)

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Let X be a random variable with finite mean $\mathbb{E}[X]$ and finite variance $\text{Var}(X)$. Then, for any fixed $\alpha > 0$,

$$\mathbb{P}\left(\left\{|X - \mathbb{E}[X]| \geq \alpha\right\}\right) \leq \frac{\text{Var}(X)}{\alpha^2}.$$



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Proof of Theorem 2:

- For any fixed $\alpha > 0$, we have

$$\begin{aligned} \mathbb{P}\left(\left\{|X - \mathbb{E}[X]| \geq \alpha\right\}\right) &= \mathbb{P}\left(\left\{(X - \mathbb{E}[X])^2 \geq \alpha^2\right\}\right) \\ &\leq \frac{\mathbb{E}\left[(X - \mathbb{E}[X])^2\right]}{\alpha^2} && \text{(Markov's inequality applied to } (X - \mathbb{E}[X])^2\text{)} \\ &= \frac{\text{Var}(X)}{\alpha^2}. \end{aligned}$$

Jensen's Inequality

Theorem 3 (Jensen's Inequality)

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a convex and differentiable function, i.e.,

$$g(y) \geq g(x) + g'(x)(y - x) \quad \forall x, y \in \mathbb{R}.$$

If $|\mathbb{E}[X]| < +\infty$ and $|\mathbb{E}[g(X)]| < +\infty$, then

$$\mathbb{E}[g(X)] \geq g(\mathbb{E}[X]).$$

Corollary:

- $\mathbb{E}[X^2] \geq (\mathbb{E}[X])^2, \quad \mathbb{E}[|X|] \geq |\mathbb{E}[X]|$
- $\mathbb{E}[\log X] \leq \log \mathbb{E}[X]$ for a non-negative random variable X

Jensen's Inequality

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If $|\mathbb{E}[X]| < +\infty$ and $|\mathbb{E}[g(X)]| < +\infty$, then

$$\mathbb{E}[g(X)] \geq g(\mathbb{E}[X]).$$

Proof of Theorem 3:

- For each $\omega \in \Omega$, we have

$$g(X(\omega)) \geq g(\mathbb{E}[X]) + g'(\mathbb{E}[X])(X(\omega) - \mathbb{E}[X]).$$

- The above relation, when written compactly, reads

$$g(X) \geq g(\mathbb{E}[X]) + g'(\mathbb{E}[X])(X - \mathbb{E}[X]).$$

- The result is obtained by taking expectations on either sides

Moment Generating Function (MGF)

Moment Generating Function (MGF)

Definition (Moment Generating Function)

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let X be a random variable.

The **moment generating function (MGF)** of X is a function $M_X : \mathbb{R} \rightarrow [0, +\infty]$ defined as

$$M_X(t) := \mathbb{E}[e^{tX}], \quad t \in \mathbb{R}.$$

The **region of convergence (ROC)** of MGF is defined as the set

$$\text{ROC}(M_X) = \left\{ t \in \mathbb{R} : M_X(t) < +\infty \right\}.$$

Examples

- Fix $\mu > 0$. If $X \sim \text{Exponential}(\mu)$, then

$$M_X(t) = \begin{cases} \frac{\mu}{\mu - t}, & t < \mu, \\ +\infty, & t \geq \mu. \end{cases}$$

- If $X \sim \mathcal{N}(0, 1)$, then

$$M_X(t) = \exp\left(\frac{t^2}{2}\right), \quad t \in \mathbb{R}.$$

- If $f_X(x) = \frac{1}{\pi} \frac{1}{1+x^2}$, $x \in \mathbb{R}$, then

$$M_X(t) = \begin{cases} 1, & t = 0, \\ +\infty, & t \neq 0. \end{cases}$$

Cramér—Chernoff Bound

Theorem 4 (Chernoff Bound)

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Let X be a random variable such that for some fixed $\varepsilon > 0$, the MGF of X satisfies

$$M_X(t) < +\infty \quad \text{for all } t \in (-\varepsilon, \varepsilon)$$

Then, for any $\alpha \in \mathbb{R}$,

$$\mathbb{P}(\{X \geq \alpha\}) \leq \frac{M_X(t)}{e^{t\alpha}} \quad \forall t > 0.$$

Optimising the RHS over $t > 0$, we get that for any $\alpha \in \mathbb{R}$,

$$\mathbb{P}(\{X \geq \alpha\}) \leq \inf_{t>0} \frac{M_X(t)}{e^{t\alpha}}.$$

Cramér—Chernoff Bound

Proof of Theorem 4:

- Keeping α fixed, notice that for any $t > 0$, we have

$$\begin{aligned}\mathbb{P}(\{X \geq \alpha\}) &= \mathbb{P}(\{tX \geq t\alpha\}) \\ &= \mathbb{P}(\{e^{tX} \geq e^{t\alpha}\}) \\ &\leq \frac{\mathbb{E}[e^{tX}]}{e^{t\alpha}} \quad (\text{Markov's inequality applied to } e^{tX}).\end{aligned}$$

- Because the RHS holds for any $t > 0$, we can optimise over $t > 0$ to get the desired result